

STAT 516: Basic Probability and its Applications

Lecture 1: Introduction to Probability. Sample spaces, events,
probability axioms

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✧ Quick Summary on Geometric Series:

Q1. Finite geometric sum

Problem:

$$S = 1 + x + x^2 + \dots + x^n$$

Solution:

- This is a finite geometric series with first term $a = 1$, ratio $r = x$, and $n + 1$ terms.
- Formula:
$$S = \frac{x^{n+1} - 1}{x - 1}, \quad x \neq 1$$
- If $x = 1$: $S = n + 1$.

Q2. Infinite geometric sum ($|x| < 1$)

Problem:

$$S = 1 + x + x^2 + x^3 + \dots$$

Solution:

- Use the finite formula and take limit as $n \rightarrow \infty$.
- Since $|x| < 1$, $x^{n+1} \rightarrow 0$.

$$S = \frac{1}{1 - x}$$

Error bound after n terms:

$$R_n = \frac{x^{n+1}}{1 - x}, \quad |R_n| \leq \frac{|x|^{n+1}}{1 - |x|}$$

Q3. Weighted sum (finite, coefficients increase)

Problem:

$$S = 1 + 2x + 3x^2 + \dots + nx^{n-1}$$

Solution (trick):

- Start with $G(x) = 1 + x + x^2 + \dots + x^n = \frac{1 - x^{n+1}}{1 - x}$.

- Differentiate:

$$S = G'(x) = 1 + 2x + 3x^2 + \dots + nx^{n-1}$$

- Formula:

$$S = \frac{1 - (n + 1)x^n + nx^{n+1}}{(1 - x)^2}, \quad x \neq 1$$

- If $x = 1$:

$$S = 1 + 2 + \dots + n = \frac{n(n + 1)}{2}$$

Variant:

$$\sum_{k=1}^n kx^k = \frac{x(1 - (n + 1)x^n + nx^{n+1})}{(1 - x)^2}$$

Q4. Infinite weighted sum ($|x| < 1$)

Problem:

$$S = 1 + 2x + 3x^2 + 4x^3 + \dots$$

Solution:

- From Q3 formula, let $n \rightarrow \infty$. For $|x| < 1$, higher terms vanish.
- Or differentiate $\frac{1}{1-x}$.

$$S = \frac{1}{(1 - x)^2}$$

Variant:

$$\sum_{k=1}^{\infty} kx^k = \frac{x}{(1 - x)^2}, \quad |x| < 1$$

✧ Differentiation of Integrals:

Q1. Differentiation of an Integral (Variable Upper Limit)

$$\frac{d}{dx} \int_a^{g(x)} f(t) dt = f(g(x)) \cdot g'(x)$$

- If $g(x) = x$, this reduces to the usual FTC Part I:

$$\frac{d}{dx} \int_a^x f(t) dt = f(x).$$

* Double Integrals and Sums:

Q1. $\int_0^\infty ye^{-y} dy$

Method: Gamma integral / integration by parts.
Let $u = y$, $dv = e^{-y} dy \Rightarrow du = dy$, $v = -e^{-y}$.

$$\int_0^\infty ye^{-y} dy = [-ye^{-y}]_0^\infty + \int_0^\infty 0^\infty e^{-y} dy = 0 + [-e^{-y}]_0^\infty = 1.$$

Answer: 1.

Q2. $\int_0^\infty \int_0^y e^{-y} dx dy$

Direct (inner integral first):

The integrand doesn't depend on x , and x runs from 0 to y , so

$$\int_0^y e^{-y} dx = ye^{-y}.$$

Hence the double integral becomes

$$\int_0^\infty ye^{-y} dy = 1 \quad (\text{from Q1}).$$

Answer: 1.

Alternate (swap order): Region is $\{(x, y) : 0 \leq x \leq y < \infty\}$.

Switching order: $0 \leq x < \infty$, $y \in [x, \infty)$,

$$\int_0^\infty \int_x^\infty e^{-y} dy dx = \int_0^\infty e^{-x} dx = 1.$$

Q4. $\int_{x=1}^e \int_{y=0}^e 1 dy dx$

Step 1. Interpret the region

- $x \in [1, e]$, $y \in [0, e]$.
- This is a rectangle of width $e - 1$ (in x) and height e (in y).

Step 2. Compute directly

$$\int_1^e \int_0^e 1 dy dx = \int_1^e [y]_0^e dx = \int_1^e 1^e dx = e(e - 1).$$

Step 3. Express using $\ln$

The interval length $[1, e]$ in the x -direction can also be written as $\ln(e) - \ln(1) = 1 - 0 = 1$.

So another way to phrase the result is

$$\int_1^e \int_0^e 1 dy dx = e \cdot (\ln e - \ln 1) \cdot (e - 1) = e(e - 1).$$

Final Answer:

$$e(e - 1) \quad (\text{equivalently written with } \ln: e(\ln e - \ln 1)(e - 1)).$$

Q3. Change order: $\int_0^\infty \int_{y=x}^\infty e^{-y} dy dx$

1) Describe the region

Original limits: $0 \leq x < \infty$ and y runs from x to ∞ .

Region $R = \{(x, y) : 0 \leq x \leq y < \infty\}$.

2) Swap the order

Fix y first: y ranges from 0 to ∞ .

For a given y , x runs from 0 up to y .

So

$$\int_0^\infty \int_x^\infty e^{-y} dy dx = \int_0^\infty \int_0^y e^{-y} dx dy.$$

3) Evaluate

Inner integral (no x -dependence):

$$\int_0^y e^{-y} dx = ye^{-y}. \quad (\text{Quick check via original order})$$

Then

$$\int_0^\infty ye^{-y} dy = 1. \quad \int_0^\infty \left(\int_x^\infty e^{-y} dy \right) dx = \int_0^\infty e^{-x} dx = 1.$$

Answer: Swapped form $\int_0^\infty \int_0^y e^{-y} dx dy$, value 1.

Q5. $\int_{y=0}^1 \int_{x=e^y}^e 1 dx dy$

Step 1. Evaluate directly

$$\int_0^1 [x]_{x=e^y}^e dy = \int_0^1 (e - e^y) dy = e \cdot (1) - [e^y]_0^1 = e - (e - 1) = 1.$$

Step 2. Change order (uses $\ln$)

Region: $0 \leq y \leq 1$, $e^y \leq x \leq e$.

Swap: $1 \leq x \leq e$, $0 \leq y \leq \ln x$.

$$\int_{x=1}^e \int_{y=0}^{\ln x} 1 dy dx = \int_1^e \ln x dx = [x \ln x - x]_1^e = (e \cdot 1 - e) - (0 - 1) = 1.$$

Final Answer: 1.

Some Limits and Tangencies:

Q1. Limit of Tangency (Definition of e)

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n$$

Solution

- This is the standard definition of the mathematical constant e .
- As n grows, $\left(1 + \frac{1}{n}\right)^n$ approaches a fixed value.

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e \approx 2.71828$$

Final Answer: e

Q2. Generalized Limit

$$\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n$$

Solution

- Recognize it as the extension of

$$\lim_{n \rightarrow \infty} \left(1 + \frac{1}{n}\right)^n = e.$$

- By definition, this converges to the exponential function:

$$\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x.$$

Final Answer: e^x

If $x=1$, answer should be same as Q1

Q3(a). Compare $1+x$ and e^x for $0 < x < 1$

Step 1. Recall the inequality

For all real x , we have

$$e^x \geq 1 + x,$$

with equality only at $x = 0$.

This comes from the fact that e^x is convex (its tangent at $x = 0$ is the line $y = 1 + x$).

Step 2. Apply to $0 < x < 1$

- At $x = 0$: $e^0 = 1$, $1 + 0 = 1$, so equal.
- For $0 < x < 1$: strict inequality holds:
 $e^x > 1 + x$.

Final Answer:

For $0 < x < 1$,

$$e^x > 1 + x$$

Q3(b). Compare $1-x$ and e^{-x}

Using the standard inequality $e^u \geq 1 + u$ for all real u (equality only at $u = 0$), set $u = -x$. Then

$$e^{-x} \geq 1 - x,$$

with equality only when $x = 0$.

- For $x > 0$ (including $0 < x < 1$): $e^{-x} > 1 - x$.
- For $x = 0$: they're equal.
- For $x < 0$: still $e^{-x} > 1 - x$ (strict, since $x \neq 0$).

Answer: e^{-x} is bigger (except at $x = 0$, where they're equal).

Q4. Harmonic Numbers

Definition:

$$H_n = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{n} = \sum_{k=1}^n \frac{1}{k}.$$

Key Facts

- Growth: $H_n \rightarrow \infty$ as $n \rightarrow \infty$, but slowly.

- Asymptotic:

$$H_n \sim \ln n + \gamma \quad (n \rightarrow \infty),$$

where $\gamma \approx 0.5772$ is the Euler–Mascheroni constant.

- Bounds:

$$\ln(n+1) \leq H_n \leq 1 + \ln n.$$

- Examples:

$$H_1 = 1, \quad H_2 = \frac{3}{2}, \quad H_3 = \frac{11}{6}, \quad H_4 = \frac{25}{12}.$$

Final Note: Harmonic numbers occur in analysis, probability, and number theory (e.g., in algorithm complexity).

Random Experiments and a Sample Space

- ▶ A **random experiment** can have several **possible outcomes**
- ▶ The **sample space** of a random experiment $\mathcal{S}$ is the **set** of all its possible outcomes.
- ▶ Example: tossing the coin once we have the space $\mathcal{S} = \{H, T\}$ *Heads, Tails*
- ▶ Example: tossing the coin twice and recording the outcomes produces $\mathcal{S} = \{HH, HT, TH, TT\}$ *list of all the possible outcomes*
- ▶ Example: roll a die with 6 faces and record the outcome. Then, $\mathcal{S} = \{1, 2, 3, 4, 5, 6\}$.

$1 \in \mathcal{S}$
 $2 \in \mathcal{S}$
 $0.5 \notin \mathcal{S}$ (not a set of $\mathcal{S}$)

Practical Applications:

- Genetics
- CS (Randomized Algorithms)
- Physics (Quantum)
- Statistics

The idea of probability

Ex): flipping a coin many times, prop of heads would approach 0.5 eventually.

- ▶ For us, **probability** is the **idealized quantity** the **relative frequency** approaches as the number of experiments grows
- ▶ Other approaches are possible...e.g. treating the probability as as personal (subjective) statement of beliefs
- ▶ The theory does not change

Basic Set theory I

- ▶ A **set** is a collection of objects
- ▶ Example: if $\mathbb{R}$ is a set of all real numbers, then 0.5 is an **element** of $\mathbb{R}$ *"an element of"*
- ▶ Notation: $0.5 \in \mathbb{R}$
- ▶ If $C = \{x : 0 \leq x \leq 1\}$, we say that $0.5 \in C$ as well
- ▶ A set C is **countable** if it can be **enumerated**

↓
If you can list its
elements in order

Examples

• Countable Set? only if:
 $\{., \# , \sim\}$ ① finite sets
② exists one to one map between that set, and
the set of natural numbers.
↳ if you can list its elements in sequence.

- ▶ Neither $\mathbb{C}$ nor $\mathbb{R}$ are countable
- ▶ The set $\{1, 2, 3, 4, 5\}$ is countable
- ▶ The set of all integers $\{\dots, -2, -1, 0, 1, 2, \dots\}$ is also countable

finite sets are "listable"

Basic Set theory II

- ▶ A is a subset of B if for every $x \in A$ we have $x \in B$



- ▶ Notation: $A \subset B$

- ▶ If $A \subset B$ and $B \subset A$, then $A = B$ *fully overlap*

- ▶ A set A is an **empty set** if it has no elements.

- ▶ Notation: $A = \emptyset$ *no outcome occurs, $\Rightarrow$ Impossible Event with probability zero.*

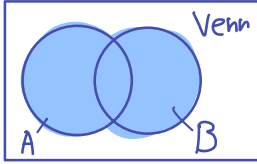
Some Relations from Set Theory

- ▶ The **union** of two events A and B is the event consisting of all outcomes that are *either in A or in B or in both events*.
Notation: $A \cup B$. Reading: A or B
- ▶ The **intersection** of two events A and B is the event consisting of all outcomes that are in *both A and B* . Notation $A \cap B$. It is read " A and B ".
- ▶ The **complement** of an event A , denoted by A' , is the set of all outcomes in S that are *not in A* .

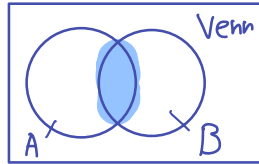
Some Relations from the Set Theory

- Each regions of A and B , except for the regions they overlap.
- ▶ The set $A \triangle B$ consists of all outcomes that are either in A or in B but not in both. Reading: a **symmetric difference** of A and B
 - ▶ The **difference** between two sets A and B or a **relative complement of B in A** $A \setminus B = A \cap B'^{\text{not}}$ consists of all outcomes in A that are *not* in B *outcomes in A but not B*
 - ▶ The notation ~~$A - B$~~ is also used but is less preferred because it can also mean the set of all $\{a - b \mid a \in A, b \in B\}$
 - ▶ As an example, $A \cup B \cup C$ means **at least one of A , B or C occurred**
 - ▶ $A \cap B \cap C$ means that **all of A , B and C occurred**
 - ▶ $A \cap B' \cap C'$ means that **A occurred but not B or C**

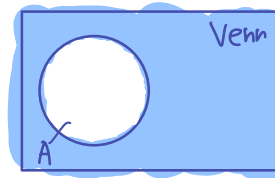
$A \cup B$ "or"



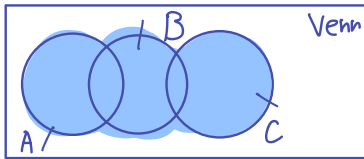
$A \cap B$ "and"



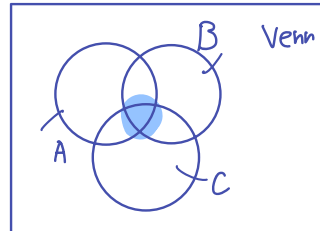
A' "not"



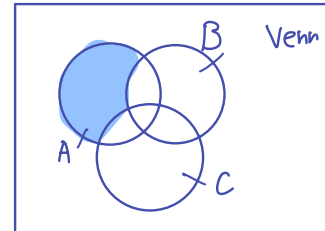
$A \cup B \cup C$



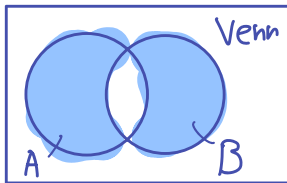
$A \cap B \cap C$



$A \cap B' \cap C'$



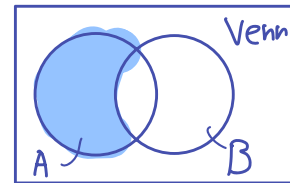
$A \Delta B$ "difference"



difference between A and B

↑

$A \setminus B = A \cap B'$



☒ $A \setminus B$ Correct

~~☒ $A = B$ Incorrect~~

Examples

- ▶ If one event is impossible ($\emptyset$), the union doesn't change outcome.
If $C_1 = \emptyset$, $C_1 \cup C_2 = C_2$ for any C_2 Ex): $C_1 = \emptyset, C_2 = \{1, 2\} \Rightarrow C_1 \cup C_2 = \{1, 2\}$
- ▶ Idempotent Law (union with itself doesn't add new elements)
For any C $C \cup C = C$
- ▶ $C_1 = \{1, 2, 3\}$ and $C_2 = \{2, 3, 4\}$
- ▶ $C_1 \cup C_2 = \{1, 2, 3, 4\}$ and $C_1 \cap C_2 = \{2, 3\}$
- ▶ Consider rolling a die with the sample space $S = \{1, 2, 3, 4, 5, 6\}$.
- ▶ $A' = \{4, 5, 6\}$.
- ▶ $A = \{1, 2, 3\}$

Some Additional Set Theory Relations

$$A_k = \{0, 1, 2, \dots, k\} \hookrightarrow \begin{aligned} A_1 &= \{0, 1\} \\ A_2 &= \{0, 1, 2\} \\ A_3 &= \{0, 1, 2, 3\} \dots \text{etc} \end{aligned}$$

- There exists at least one index k_0 such that $x \in A_{k_0}$
- x belongs to at least one of the sets.

► For a sample space $\mathcal{S}$, an element $x \in \bigcup_k A_k$ if and only if there exists k_0 such that $x \in A_{k_0}$.

Everything

► $x \in \bigcap_k A_k$ if and only if $x \in A_k$ for all $k \rightarrow$ x is in every single set of A_k
 x belongs to all of the sets simultaneously.

Union Across Infinite Sets ($\cup$)

Example: if $A_k = \{0, 1, 2, \dots, k\}$ then $\bigcup_k A_k = \{0, 1, 2, 3, \dots\}$
 and $\bigcap_k A_k = \{0, 1\}$ Intersection Across Infinite Sets ($\cap$)

► Example: let $C_k = \left\{x : \frac{1}{k+1} \leq x \leq 1\right\}$, $k = 1, 2, 3, \dots$. Then,

$$\bigcup_{k=1}^{\infty} C_k = \{x : 0 \leq x \leq 1\}$$

Continuous Sets

► Example: let $C_k = \left\{x : 0 < x < \frac{1}{k}\right\}$ for $k = 1, 2, 3, \dots$. Then,

$$\bigcap_{k=1}^{\infty} C_k = \emptyset.$$

$$C_1 = (0, 1)$$

$$C_2 = (0, 0.5)$$

$$C_3 = (0, 0.333)$$

◦ However, 0 is excluded in $0 < x$,
 so intersection is $\emptyset$.

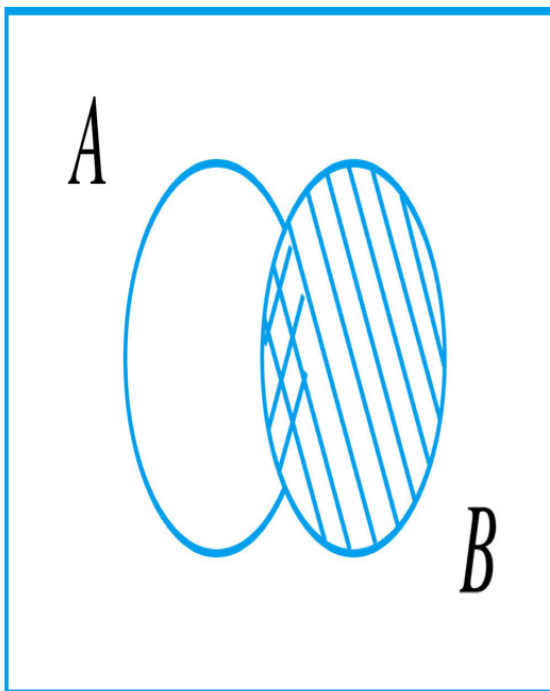
$$C_k = \left\{x : \frac{1}{k+1} \leq x \leq 1\right\} \hookrightarrow C_1 = \left\{x : \frac{1}{2} \leq x \leq 1\right\} = [0.5, 1]$$

left end-point is moving closer to zero.

$$C_2 = \left[\frac{1}{3}, 1\right]$$

$$C_3 = \left[\frac{1}{4}, 1\right] \dots \text{etc}$$

Venn Diagrams



Core Notations (Lecture 1)

1. Element and Set Membership

- $x \in A$: x is in set A .
- $x \notin A$: x is not in set A .

2. Subset

- $A \subset B$: All elements of A are in B .
- Example: $\{1, 2\} \subset \{1, 2, 3, 4\}$.

3. Empty Set

- $\emptyset$: the set with no elements (impossible event).

4. Union

- $A \cup B$: all outcomes in A or B (or both).
- Think: "A or B".

5. Intersection

- $A \cap B$: all outcomes in both A and B .
- Think: "A and B".

6. Complement

- A' : all outcomes not in A .
- If $A = \{1, 2, 3\}$ and $S = \{1, 2, 3, 4, 5, 6\}$, then $A' = \{4, 5, 6\}$.

7. Difference (Relative Complement)

- $A \setminus B = A \cap B'$: all outcomes in A but not in B .
- Example: $\{1, 2, 3, 4\} \setminus \{3, 4, 5, 6\} = \{1, 2\}$.

8. Symmetric Difference

- $A \Delta B$: outcomes in A or B , but not both.

9. Infinite Unions/Intersections

- $\bigcup_{k=1}^{\infty} A_k$: elements that appear in **at least one** A_k .
- $\bigcap_{k=1}^{\infty} A_k$: elements that appear in **all** A_k .

Some operation laws on sets

► Distributive laws:

$$B \cap (\cup_k A_k) = \cup_k (B \cap A_k)$$

any set of A
Ex) $A_1, A_2, \dots$

• Example:

• If $B = \{1, 2\}$, $A_1 = \{2, 3\}$, $A_2 = \{3, 4\}$:

• Left-hand side: $B \cap (A_1 \cup A_2) = \{1, 2\} \cap \{2, 3, 4\} = \{2\}$.

• Right-hand side: $(B \cap A_1) \cup (B \cap A_2) = \{2\} \cup \emptyset = \{2\}$. ✓ Works.

$$B \cup (\cap_k A_k) = \cap_k (B \cup A_k)$$

► De Morgan's laws: for any two sets A_1 and A_2

"Not (A and B)" = "Not A or Not B"

$$[A_1 \cap A_2]' = A_1' \cup A_2'$$

• Example with dice:

• Let $A_1 = \{1, 2, 3\}$, $A_2 = \{3, 4, 5\}$, $S = \{1, 2, 3, 4, 5, 6\}$.

• Then $(A_1 \cap A_2)' = (\{3\})' = \{1, 2, 4, 5, 6\}$.

• Meanwhile, $A_1' \cup A_2' = \{4, 5, 6\} \cup \{1, 2, 6\} = \{1, 2, 4, 5, 6\}$. ✓ Matches.

"Not (A or B)" = "Not A and Not B"

$$[A_1 \cup A_2]' = A_1' \cap A_2'$$